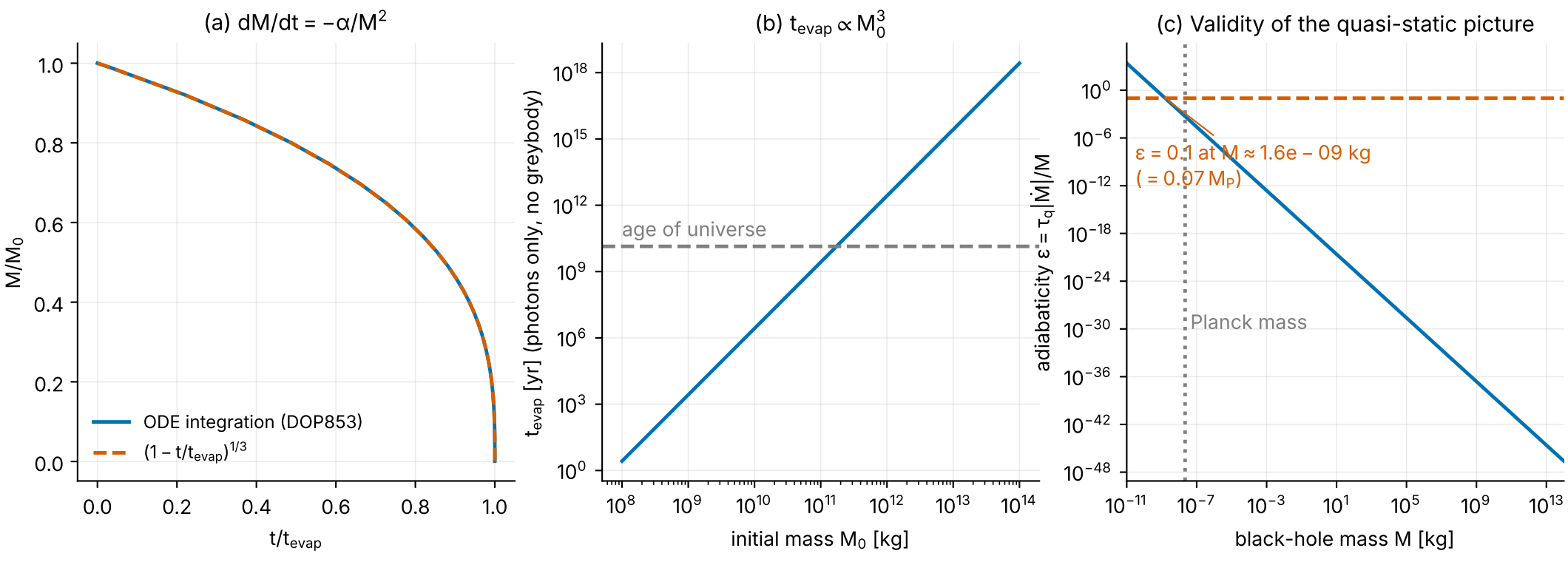
(a) dM/dt = −α/M²
M/M₀
t/t_evap
ODE integration (DOP853)
(1 − t/t_evap)^1/3
(b) t_evap ∝ M₀³
t_evap [yr] (photons only, no greybody)
age of universe
initial mass M₀ [kg]
(c) Validity of the quasi-static picture
adiabaticity ε = τ_q|Ṁ|/M
ε = 0.1 at M ≈ 1.6e − 09 kg
(= 0.07 M_P)
Planck mass
black-hole mass M [kg]